Supplementary Material

Point-Selection Fine-Tuning Framework for Robust Point Cloud Classification

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S1 Analysis of Augmentation Results

Tables S1, S2, and S3 show dataset-dependent behavior. On ModelNet-C [9] and ModelNet40-C [11], PSFT with WOLFMix [9] achieves state-of-the-art corruption robustness among all compared settings. With Uni3D-B [18], it reaches the best mCE of 0.465 on ModelNet-C and the best mER of 12.6 on ModelNet40-C. These results indicate that augmentation and lightweight robust adaptation are highly complementary on widely used 3D corruption benchmarks [9,11,14], while maintaining competitive clean OA.

On ScanObjectNN-C [14], WOLFMix is inconsistent and can hurt some backbones. Unlike synthetic benchmarks, ScanObjectNN contains real scans with heavier clutter, occlusion, and partial observations [13], making its corruption benchmark harder to optimize [14]. Combining strong augmentation with lightweight fine-tuning may therefore destabilize convergence and weaken robustness.

Table S1. ModelNet-C results with augmentation. OA is overall accuracy; mCE is mean corruption error; all other columns report CE (↓). W.M. denotes WOLFMix [9]; G/L indicate global/local corruptions. Other settings are from [2,3,5]; bold marks column-best values.

Backbone	Setting	OA (%) ↑	mCE ↓	Corruption CE (↓)						
				Scale	Jitter	Drop-G	Drop-L	Add-G	Add-L	Rotate
GDANet [15]	Baseline	93.4	0.892	0.830	0.839	0.794	0.894	0.871	1.036	0.981
	W.M.	**93.4**	0.571	0.904	0.883	0.532	0.551	0.305	0.415	0.409
	FAT + W.M.	93.0	0.537	1.138	**0.418**	0.460	0.527	0.281	0.404	0.530
RPC [8]	Baseline	93.0	0.863	0.840	0.892	0.492	0.797	0.929	1.011	1.079
	AdaptPoint++	–	0.555	0.957	0.759	0.367	0.705	0.258	**0.273**	0.563
	EPiC + W.M.	92.7	0.501	0.915	0.680	0.315	**0.420**	0.251	0.382	0.544
Point-BERT [17]	Baseline	92.2	1.098	0.872	1.503	0.512	0.903	1.085	1.495	1.316
	PSFT (ours)	91.4	0.790	1.149	0.503	0.383	0.961	0.319	0.625	1.591
	PSFT + W.M.	90.4	0.617	1.245	0.440	0.415	0.696	0.329	0.425	0.772
Point-MAE [6]	Baseline	92.6	0.983	0.840	1.256	0.464	0.763	1.075	1.265	1.219
	PSFT (ours)	91.9	0.704	1.032	0.427	0.431	0.860	0.298	0.502	1.377
	PSFT + W.M.	89.8	0.630	1.255	0.475	0.480	0.696	0.366	0.404	0.735
ULIP-2 [16]	Baseline	92.1	0.863	0.851	1.778	0.492	0.734	0.386	0.862	0.935
	PSFT (ours)	93.3	0.530	0.819	0.585	0.298	0.816	**0.244**	0.567	0.381
	PSFT + W.M.	93.2	0.473	0.851	0.665	**0.294**	0.517	0.251	0.396	**0.340**
Uni3D-B [18]	Baseline	92.9	0.746	**0.777**	1.228	0.415	0.787	0.444	0.833	0.740
	PSFT (ours)	93.4	0.566	0.862	0.459	0.327	0.986	0.254	0.575	0.498
	PSFT + W.M.	93.0	**0.465**	0.894	0.506	0.327	0.575	0.254	0.356	0.344

Table S2. ModelNet40-C error rates with augmentation. mER is mean error rate; W.M. denotes WOLFMix [9]. Dens., Up., Bg., Rot., and iRBF abbreviate density, upsampling, background, rotation, and inverse RBF. Corruptions are grouped by type within one table; lower is better, and bold marks column-best values.

Density corruptions							
Backbone	Setting	mER ↓	Occlusion	LiDAR	Dens.+	Dens.-	Cutout
PE [12]	Baseline	22.1	–	–	–	–	–
OmniVec2 [10]	Baseline	14.2	–	–	–	–	–
Point-BERT [17]	Baseline	26.5	54.7	73.8	10.8	13.4	13.6
	PSFT (ours)	20.0	56.9	70.9	9.2	10.6	9.6
	PSFT + W.M.	16.9	56.3	60.0	10.2	10.1	9.8
Point-MAE [6]	Baseline	23.9	53.5	74.8	10.7	11.3	11.9
	PSFT (ours)	18.3	55.5	66.6	9.0	10.7	10.1
	PSFT + W.M.	16.9	55.3	58.4	10.5	10.7	10.4
ULIP-2 [16]	Baseline	24.1	51.4	80.0	9.7	10.8	10.9
	PSFT (ours)	14.9	44.6	53.3	7.5	8.8	9.0
	PSFT + W.M.	13.9	40.8	51.9	**7.0**	**7.6**	**7.5**
Uni3D-B [18]	Baseline	18.9	44.2	58.2	9.6	10.3	11.3
	PSFT (ours)	14.2	41.2	**42.4**	8.1	9.8	10.5
	PSFT + W.M.	**12.6**	**40.1**	42.6	7.5	7.7	7.7
Noise corruptions							
Backbone	Setting	mER ↓	Uniform	Gaussian	Impulse	Up.	Bg.
PE [12]	Baseline	22.1	14.1	17.1	15.7	13.7	17.9
OmniVec2 [10]	Baseline	14.2	–	–	–	–	–
Point-BERT [17]	Baseline	26.5	17.5	22.8	46.7	23.8	36.7
	PSFT (ours)	20.0	11.1	11.2	10.4	10.2	9.9
	PSFT + W.M.	16.9	11.3	11.2	9.8	10.7	10.5
Point-MAE [6]	Baseline	23.9	15.7	18.5	33.0	18.8	35.5
	PSFT (ours)	18.3	10.5	**10.3**	9.4	9.6	9.6
	PSFT + W.M.	16.9	11.9	11.6	9.7	10.7	11.2
ULIP-2 [16]	Baseline	24.1	29.9	35.7	18.7	31.9	12.9
	PSFT (ours)	14.9	12.0	12.8	8.2	9.5	7.6
	PSFT + W.M.	13.9	14.9	15.8	**7.7**	10.4	**7.5**
Uni3D-B [18]	Baseline	18.9	15.1	18.9	20.2	21.6	14.1
	PSFT (ours)	14.2	**10.0**	10.8	8.9	9.7	7.6
	PSFT + W.M.	**12.6**	10.8	11.1	7.8	**9.5**	8.0
Transformation corruptions							
Backbone	Setting	mER ↓	Rotation	Shear	FFD	RBF	iRBF
PE [12]	Baseline	22.1	–	–	–	–	–
OmniVec2 [10]	Baseline	14.2	–	–	–	–	–
Point-BERT [17]	Baseline	26.5	25.3	13.3	14.5	15.9	14.9
	PSFT (ours)	20.0	30.7	16.9	15.5	14.1	13.5
	PSFT + W.M.	16.9	12.6	10.2	10.3	10.0	10.0
Point-MAE [6]	Baseline	23.9	22.7	11.5	13.1	13.9	13.0
	PSFT (ours)	18.3	24.3	12.9	12.9	12.0	11.3
	PSFT + W.M.	16.9	11.7	10.5	10.2	10.3	10.0
ULIP-2 [16]	Baseline	24.1	18.6	10.5	12.1	14.7	13.8
	PSFT (ours)	14.9	7.6	7.1	10.4	12.5	11.9
	PSFT + W.M.	13.9	7.2	**6.8**	7.3	7.7	7.8
Uni3D-B [18]	Baseline	18.9	13.4	8.8	11.7	13.4	12.6
	PSFT (ours)	14.2	9.0	7.5	11.1	13.3	12.7
	PSFT + W.M.	**12.6**	**7.0**	7.1	**7.3**	**7.7**	**7.7**

Table S3. ScanObjectNN-C results with augmentation. OA and mCE denote overall accuracy and mean corruption error. W.M. denotes WOLFMix [9]; PointAugment [4] and PointMixup [1] are additional augmentations. G and L denote global and local corruptions. Bold denotes the best result in each column.

Backbone	Setting	OA (%) ↑	mCE ↓	Corruption CE (↓)						
				Scale	Jitter	Drop-G	Drop-L	Add-G	Add-L	Rotate
PointNeXt [7]	Baseline	87.3	0.921	0.803	1.079	0.807	0.942	0.944	0.875	0.995
	W.M.	**87.7**	0.869	0.819	1.193	0.897	0.780	0.893	**0.800**	0.700
	PointAugment	87.4	0.894	0.808	1.000	0.775	0.858	0.976	0.850	0.993
	PointMixup	88.2	0.904	0.801	1.035	0.799	0.927	0.876	0.837	1.056
Point-BERT [17]	Baseline	83.4	0.931	0.618	1.088	0.587	0.736	0.661	1.630	1.195
	PSFT (ours)	79.4	0.970	0.898	0.954	0.608	0.937	0.493	1.612	1.288
	PSFT + W.M.	65.6	1.235	0.711	1.217	0.931	1.327	0.793	2.075	1.592
Point-MAE [6]	Baseline	84.3	0.895	0.609	0.952	0.569	0.726	0.550	1.722	1.135
	PSFT (ours)	77.6	1.014	0.960	0.974	0.701	0.990	0.580	1.568	1.326
	PSFT + W.M.	64.3	1.333	0.727	1.344	0.997	1.389	0.876	2.291	1.704
ULIP-2 [16]	Baseline	86.2	0.777	0.538	1.096	0.524	**0.637**	0.376	1.348	0.918
	PSFT (ours)	86.1	**0.685**	0.621	0.960	**0.431**	0.670	**0.333**	1.079	**0.700**
	PSFT + W.M.	82.3	0.786	**0.412**	1.189	0.585	0.842	0.424	1.278	0.775
Uni3D-B [18]	Baseline	85.8	0.801	0.509	1.085	0.532	0.680	0.365	1.502	0.933
	PSFT (ours)	86.4	0.702	0.626	**0.871**	0.458	0.733	0.341	1.079	0.805
	PSFT + W.M.	80.5	0.833	0.448	1.094	0.622	0.947	0.435	1.427	0.861

These results quantify the dataset dependence. Averaged over four backbones, WOLFMix lowers ModelNet-C mCE by 0.101 and ModelNet40-C mER by 1.8 points, but raises ScanObjectNN-C mCE by 0.204 and reduces clean OA by 9.2 points. All backbones follow this negative ScanObjectNN-C trend, establishing an augmentation–dataset mismatch whose cause–clutter, occlusion, partial observations, or interaction with scan noise–requires controlled study.

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